PROCODE

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RESPONSIBILITY

NUPUL : UX

DIVESH : WEBSITE PLANNER

Shrutesh : BACKEND

Swapnil : FRONTEND

AI-powered Brand Governance Platform

1. Understanding of the Given Problem Statement

The problem statement provided by Inkloom AI highlights the difficulty organisations face in maintaining a consistent, accurate, and trustworthy brand identity when content is created by multiple employees, agencies, and AI tools.

Different content creators may unintentionally use inconsistent language, incorrect facts, outdated statistics, inappropriate brand tones, unapproved visuals, or unsupported promotional claims. These mistakes can damage an organisation's credibility, confuse customers, and reduce trust in its communications.

Our understanding is that the problem is not simply about identifying spelling or grammatical errors. It is about ensuring that every piece of content follows the organisation's approved brand guidelines, uses reliable information, and maintains consistent messaging before publication.

To address this problem, we propose an AI-powered Brand Governance Platform that analyses text, documents, and supported images against approved brand rules, facts, terminology, and visual assets. It identifies potential violations, explains why they were detected, provides evidence and correction suggestions, and allows authorised reviewers to approve or request changes.

The platform will also help organisations identify outdated information, resolve conflicting brand rules, maintain version history, and recognise recurring content-quality issues. AI will assist with analysis and recommendations, while final decisions remain under human control.

Our primary objective is to prevent brand inconsistencies before content is published, helping organisations maintain credibility, improve content quality, and establish a reliable and transparent content-review process.

2. Users Affected by the Problem

The problem affects individuals and organisations that create, review, manage, or publish content. When multiple people and AI tools produce content without a shared system for checking brand guidelines, inconsistencies, outdated information, and unsupported claims can go unnoticed. Brand governance becomes particularly important for teams working across different departments, agencies, languages, and communication channels.

Primary and secondary users

1. Researchers

Researchers preparing reports, presentations, publications, and technical documents need accurate facts, consistent terminology, and reliable references. Outdated data or unsupported claims can reduce the credibility of their work.

2. Educational Professionals

Teachers, educational institutions, and academic organisations produce notices, course materials, presentations, and digital content. Inconsistent information or outdated details can confuse students and other stakeholders.

3. Students

Students creating presentations, academic projects, portfolios, and other digital content can benefit from guidance on consistent language, reliable information, and appropriate visual assets.

4. Large Organisations and Enterprises

Large organisations often have multiple departments, employees, regional offices, and external agencies producing content. Maintaining a consistent brand identity across these contributors can become difficult without centralised guidelines, clear approval responsibilities, and version control.

5. General Users and Independent Content Creators

Individuals, freelancers, and independent creators may lack dedicated brand-management teams. They need accessible tools to check their content, identify potential errors, and improve consistency before sharing or publishing it.

6. Industry Workers

Employees preparing technical documents, product descriptions, internal reports, and customer-facing materials need to ensure that information is current, terminology is consistent, and approved communication standards are followed.

7. Digital Marketing Agencies

Marketing agencies create content for multiple clients, each with different brand guidelines, preferred terminology, visual identities, and approval requirements. A centralised checking system can help them reduce cross-client inconsistencies and manage revisions more systematically.

8. Brand Managers, Editors, and Authorised Reviewers

These users are responsible for checking content, resolving conflicting rules, verifying claims, and approving publication. Manual review can become time-consuming when content volume increases or when several contributors require approval.

Who is most directly affected?

For our proposed solution, **the primary target users will be digital marketing agencies, brand managers, content creators, and organisations with multiple content contributors**

3. Our Proposed Solution:

We propose developing an **AI-Powered Brand Governance Platform** that helps organisations maintain brand consistency, factual accuracy, and trustworthy communication across content created by employees, marketing agencies, and AI tools.

The platform will provide a centralised system where organisations can upload their brand guidelines, maintain approved facts and terminology, and review content before publication. Instead of merely rejecting content that violates brand rules, the system will explain the detected issues, identify relevant evidence, and suggest corrections while preserving the creator's original intent.

Our solution will combine AI-powered analysis with rule-based validation and human approval to make content review more reliable, transparent, and efficient.

4. Complete Product Workflow

The proposed platform will follow a structured workflow that checks content against approved brand guidelines, verifies factual accuracy, identifies potential risks, and enables human approval before publication.

Step 1: Brand Knowledge Setup

The organisation uploads its brand guideline PDF or provides a website link. The system extracts relevant rules, approved terminology, brand tone, visual guidelines, and restricted claims.

Step 2: Create a Trusted Brand Knowledge Base

The platform stores approved facts, current statistics, preferred and prohibited words, approved assets, source references, and expiry dates. Conflicting rules are flagged for authorised clarification.

Step 3: Submit Content for Review

A content creator uploads or pastes text, marketing copy, a document, or a supported image. The platform identifies the content type and selects the appropriate checks.

Step 4: Intelligent Brand Compliance Analysis

The system checks tone, terminology, translation accuracy, factual claims, outdated information, visual consistency, and applicable accessibility requirements. It links findings to relevant rules or approved sources.

Step 5: Generate an Explainable Quality Report

The platform displays a confidence indicator, detected issues, severity, supporting evidence, and suggested corrections. Uncertain findings are labelled for further verification rather than presented as confirmed errors.

Step 6: Human Review and Approval

The creator revises the content and submits it to a manager or authorised reviewer. Reviewers can approve, reject, request changes, add comments, or resolve conflicting rules.

Step 7: Finalise and Maintain Content

The platform saves the approved version, revision history, evidence, and approval record. Outdated facts and content can be flagged for updating or archived after authorised review.

Expected result: Content is checked before publication, corrections remain under human control, and the organisation maintains a traceable record of important decisions.

This workflow follows the problem statement's requirements for explainable findings, approved source material, human review, and revision history.

5. Main Features We Intend to Build

The development will prioritise a functional minimum viable product (MVP) rather than attempting to implement every possible enterprise feature during the hackathon.

1. Smart Rule Uploader	Extracts brand guidelines from uploaded PDFs and supported website content.
2. Living Fact Sheet	Stores approved facts, statistics, source references, last-reviewed dates, and expiry dates.
3. Approved and Banned Word Dictionary	Identifies prohibited terminology and recommends preferred alternatives, such as replacing “cheap” with “budget-friendly” when appropriate.
4. Brand Tone Identifier	Evaluates whether content matches the organisation's intended tone and messaging.
5. Cross-Language Brand Guardian	Detects possible translation errors, terminology inconsistencies, and changes in meaning across supported languages.
6. Content Expiry Radar	Flags expired or potentially outdated facts and content that needs verification or updating.
7. Image and Visual Compliance Checker	Checks supported images for approved asset usage, logo consistency, visual rules, and basic accessibility issues.
8. Confidence and Evidence Report	Shows confidence indicators, detected problems, evidence, applicable rules, and explanations for each finding.
9. Correction Suggestions	Recommends revised wording while preserving the creator's original intention.
10. Conflicting Rules Resolver	Identifies contradictory brand rules, explains the conflict, and requests clarification from an authorised reviewer.
11. Human Review and Approval System	Supports review comments, change requests, approval status, and authorised final decisions.
12. Version History and Audit Trail	Records revisions, findings, approvals, and relevant timestamps.
13. Governance Insights Dashboard	Summarises recurring errors, frequently violated rules, and areas where brand guidelines require clarification.

6. Technology Stack We Plan to Use

We propose a web-based architecture with a frontend for content submission and review, a backend for processing and governance rules, a database for trusted brand information, and an AI layer for contextual analysis.

Frontend — Next.js, React and TypeScript

Dashboard, file upload, review reports, correction interface and approval workflow.

Backend — Python and GeminiAPi

API endpoints, document processing, rule validation, permissions and workflow management.

AI analysis

LLM API, structured outputs, retrieval from approved brand sources, tone and semantic checks.

Data storage

PostgreSQL for guidelines, facts, users, findings, approvals and version history.

Validation, security and deployment

Deterministic rule checks, authentication, role-based permissions, secure configuration, testing and web deployment.

Technology	Purpose
Next.js + React + TypeScript	Responsive web interface and dashboard.
Python + GeminiAPI	Backend APIs and content-analysis orchestration.
PostgreSQL	Persistent storage for brand rules, facts, reviews, and audit records.
LLM API	Semantic analysis, explanations, tone evaluation, and correction suggestions.
PDF and document parsing libraries	Extract text and rules from uploaded guideline documents.
OCR or vision-capable model	Analyse supported images and extract visual or textual information.
Rule-based validation	Exact banned-word checks, expiry-date comparisons, required terminology, and deterministic constraints.
GitHub	Source-code management, collaboration, and development history.
Vercel or an equivalent hosting platform	Frontend deployment; backend hosting will be selected based on compatibility and available free resources.

8. Development Plan Across All Three Rounds

Round 1

THE PLAN

Objective: Establish the problem, solution and feasible architecture.

- Finalise the problem definition, target users, and value proposition.
- Gather supporting evidence through relevant research, interviews, or observations.
- Finalise the product workflow and prioritise MVP features.
- Decide the technology stack, system architecture, and data sources.
- Allocate team responsibilities and define privacy and human-approval safeguards.
- Prepare the submission PDF, prototype design, and development milestones.

Deliverable: A complete and feasible product plan with a clear technical approach.

Round 2

THE BUILD

Objective: Build and demonstrate the end-to-end MVP.

- Develop the dashboard and content-submission interface.
- Implement guideline upload and structured brand-rule storage.
- Build the core text checker using rule-based validation and AI analysis.
- Display detected issues, explanations, evidence, and correction suggestions.
- Implement a basic human-review and approval workflow.
- Connect the frontend, backend, AI service, and database.
- Test realistic examples, record limitations, and publish a judge-accessible demo and repository.
- Identify three specific improvements to deliver in Round 3.

Deliverable: A functional prototype demonstrating content submission, compliance analysis, correction, and review.

Round 3

THE FINISH

Objective: Improve reliability, usability and completeness.

- Fix defects identified during testing and judging.
- Deliver three clearly documented improvements identified in Round 2.
- Improve explanations, error handling, evidence traceability, and confidence indicators.
- Refine the interface and test the approval process with different user roles.
- Add selected extensions such as content expiry alerts, image checks, or conflicting-rule resolution where feasible.
- Review authentication, access permissions, confidential-data handling, and accessibility.
- Complete the source-code repository, setup instructions, architecture documentation, known limitations, and future roadmap.
- Prepare the final demonstration and ensure each team member can explain their contribution.

Deliverable: A tested, documented, demonstrable final product with a realistic roadmap.

9. Expected Impact of Our Solution

Our platform aims to help organisations publish consistent, accurate, and trustworthy content while reducing the burden of manually checking every item against separate guidelines and documents.

- Improved brand consistency: Helps ensure that employees, agencies, and AI tools use approved terminology, tone, messaging, and visual assets.
- Reduced outdated information: Flags facts, statistics, and content that have passed their review or expiry dates.
- Better factual reliability: Identifies unsupported or conflicting claims and directs creators towards approved sources.
- More efficient content review: Automates repeatable checks so reviewers can focus on contextual, sensitive, or high-risk decisions.
- Greater transparency and accountability: Maintains evidence, comments, approvals, and revision history for important content decisions.

- Improved collaboration: Creates a shared process for content creators, marketing teams, managers, and authorised reviewers.
- Better multilingual communication: Helps identify translation inconsistencies and potential changes in meaning across supported languages.
- Continuous improvement: Uses recurring findings to identify unclear rules and areas where teams need additional guidance.

How we will measure impact

Rather than claiming unverified results, we will evaluate the prototype using a set of practical metrics:

Metric	How it will be evaluated
Issue detection	Compare findings against a manually labelled test set of content examples.
Evidence quality	Check whether findings reference the correct approved rule or source.
Review efficiency	Compare the time taken to review the same examples manually and with the platform.
Correction usefulness	Ask reviewers to assess whether suggested corrections preserve meaning and meet brand rules.
Workflow completion	Measure whether a content item can move from submission through analysis to recorded human approval.
Content freshness	Test whether expired facts and overdue review dates are flagged correctly.

The expected impact is improved content quality and more consistent governance. Actual improvements in accuracy, review time, or cost will be established through testing rather than assumed in advance.

10. Future Scope of the Product After the Event

After the hackathon, we plan to evolve the prototype into a scalable brand-governance platform that supports more content formats, deeper integrations, stronger security, and organisation-specific workflows.

1. Broader Content Format Support

Extend checks to presentations, spreadsheets, video subtitles, audio transcripts, social media creatives, advertisements, and larger document collections.

2. Advanced Multilingual Brand Governance

Support more languages, terminology dictionaries, regional messaging rules, and contextual translation comparisons while preserving the intended meaning.

3. Integration With Existing Workflows

Explore integrations with content-management systems, document platforms, design tools, and marketing systems so content can be checked closer to the point of publication.

4. Advanced Governance Analytics

Add trend analysis, recurring-error reports, team-level insights, and dashboards that reveal which guidelines are frequently misunderstood or violated.

5. Enterprise-Grade Security

Introduce stronger access controls, organisation-level data isolation, configurable retention policies, secure integrations, and deployment options for organisations with confidential materials.

6. Configurable Approval and Conflict Resolution

Introduce risk-based approval paths, multiple reviewer levels, policy ownership, escalation deadlines, and a structured process for resolving contradictory brand rules.

7. Continuous Learning From Approved Feedback

Use authorised reviewer feedback to refine rule definitions, correction quality, and recurring-issue detection. Changes to brand policies would still require authorised approval rather than being silently adopted by AI.